

FRACTIONS MADE FUN: DIVIDING FRACTIONS

Let's learn how to divide fractions step-by-step!



WHAT ARE FRACTIONS?

A fraction is a part of a whole! When we divide something into equal parts, each part is a fraction. For example, if you cut a pizza into 4 equal slices and eat 1 slice, you ate $\frac{1}{4}$ of the pizza!



WHAT IS DIVISION?

Division means splitting into equal parts or sharing equally among groups.

Example: $6 \div 2 = 3$

If you have 6 cookies and share them equally between 2 friends, each friend gets 3 cookies!



REAL-LIFE FRACTION DIVISION

Dividing fractions helps us share things like pizza slices or measure ingredients for cooking. It's a skill we use every day!



QUICK FRACTION REVIEW

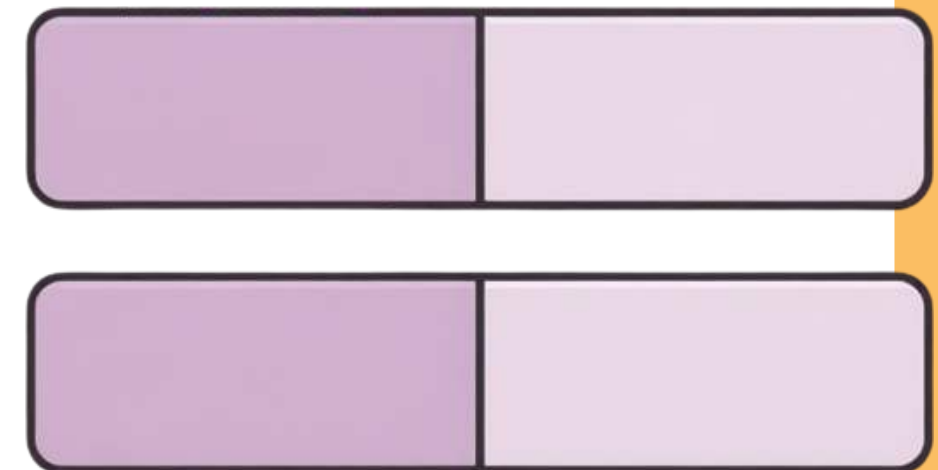
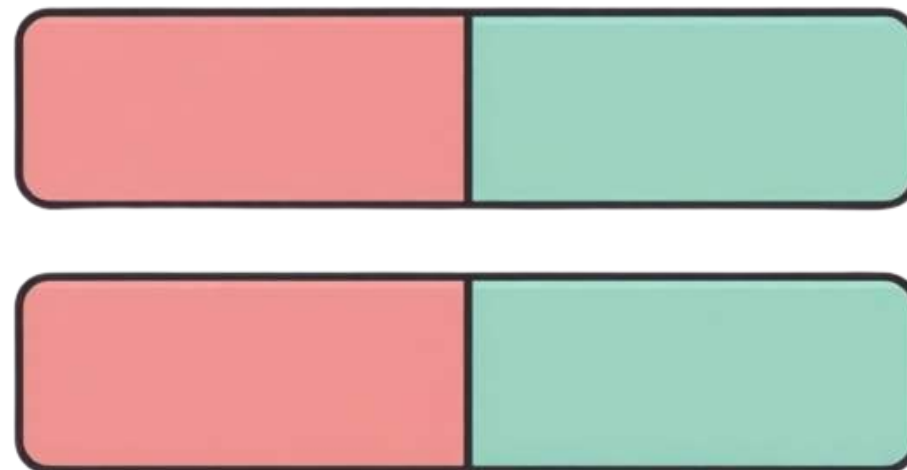
Before we divide fractions, let's remember these important basics:

- Numerator: The top number (how many parts we have)
- Denominator: The bottom number (total equal parts)
- Simplifying: Making fractions smaller ($6/8 = 3/4$)
- Proper fractions: Numerator is smaller ($2/5$)
- Improper fractions: Numerator is bigger ($7/4$)
- Mixed numbers: Whole number + fraction ($1\ 3/4$)



WHAT DOES DIVIDING FRACTIONS MEAN?

Dividing fractions means finding out how many times one fraction fits into another. Think of it like asking: "How many halves fit into three quarters?"



KEEP THE FIRST FRACTION

Keep the first fraction exactly as it is – don't change anything!

Example: $3/4 \div 1/2 \rightarrow$ Keep $3/4$



CHANGE DIVISION TO MULTIPLICATION

Change the division sign ($\div$) to a multiplication sign ($\times$).

Example: $3/4 \div 1/2$ becomes $3/4 \times ?$

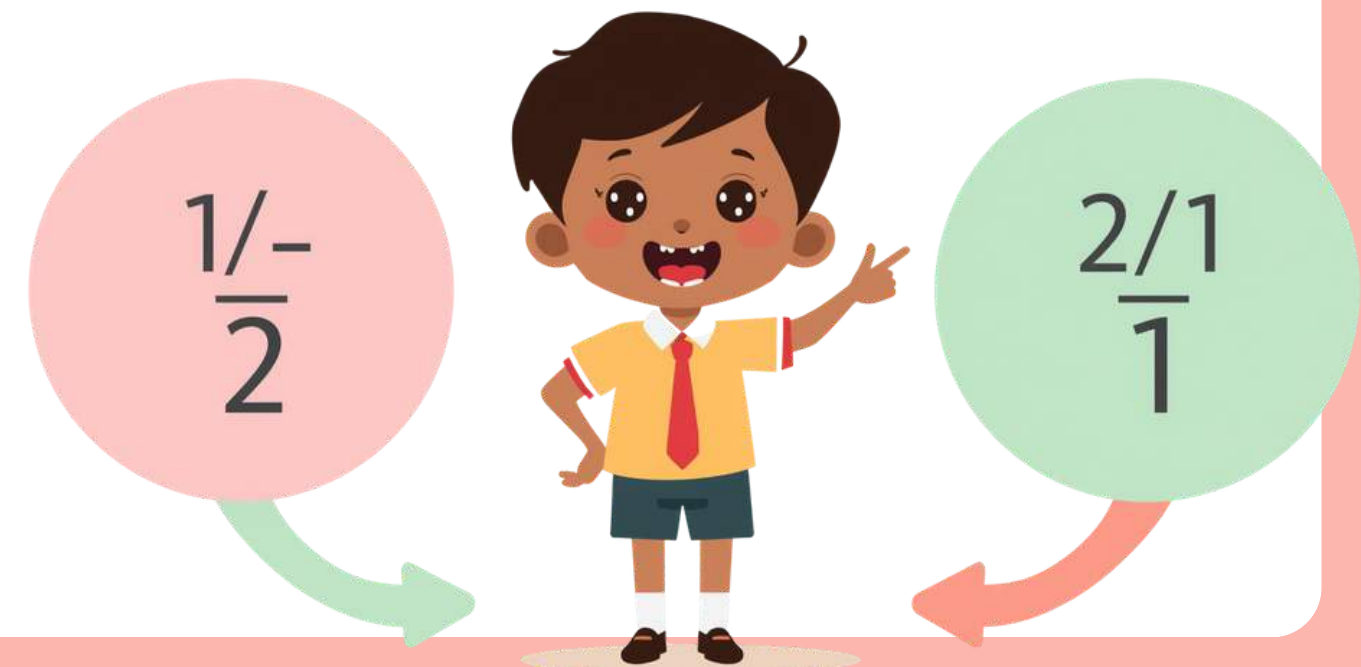


FLIP THE SECOND FRACTION

Flip the second fraction by swapping the numerator and denominator.

This is called finding the reciprocal!

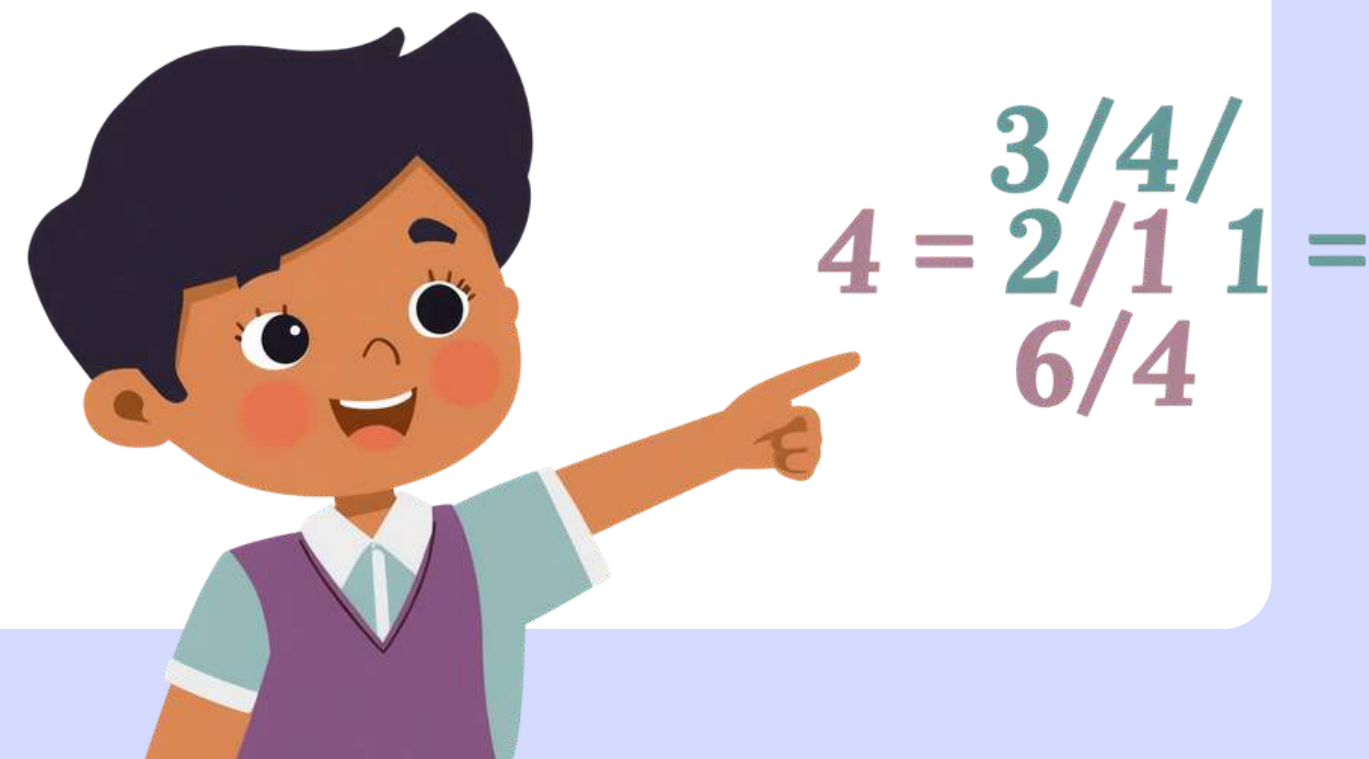
Example: $\frac{1}{2}$ flipped becomes $\frac{2}{1}$



MULTIPLY THE FRACTIONS

Multiply straight across:
numerator $\times$ numerator,
denominator $\times$ denominator.

Example: $\frac{3}{4} \times \frac{2}{1} = \frac{6}{4}$
($3 \times 2 = 6$ and $4 \times 1 = 4$)



SIMPLIFY THE RESULT

Simplify the fraction if needed. Reduce to the lowest terms by dividing numerator and denominator by their greatest common factor.

Example: $6/4 = 3/2 = 1\frac{1}{2}$ as a mixed number



VISUAL PRACTICE EXAMPLE

Let's solve this problem together!

Problem: $\frac{2}{3} \div \frac{4}{5}$

Step 1: Keep $\frac{2}{3}$

Step 2: Change $\div$ to $\times$

Step 3: Flip $\frac{4}{5}$ to get $\frac{5}{4}$

Step 4: Multiply: $\frac{2}{3} \times \frac{5}{4} = \frac{10}{12}$

Step 5: Simplify: $\frac{10}{12} = \frac{5}{6}$

Answer: $\frac{5}{6}$



WORD PROBLEMS WITH DIVIDING FRACTIONS

1. Chocolate Bars: Priya has $\frac{3}{4}$ of a chocolate bar. She wants to share it equally among 3 friends. How much does each friend get?
2. Recipe Ingredients: Arjun needs $\frac{2}{3}$ cup of flour. His measuring cup holds $\frac{1}{6}$ cup. How many scoops does he need?
3. Ribbon Pieces: Meera has $\frac{4}{5}$ metre of ribbon. She cuts it into pieces that are $\frac{1}{10}$ metre each. How many pieces can she make?



WATCH OUT!

COMMON MISTAKES TO AVOID

- ✗ Forgetting to flip the second fraction
- ✗ Multiplying instead of dividing correctly
- ✗ Not simplifying your final answer

Remember: Keep, Change, Flip, Multiply,
Simplify!



PRACTICE PROBLEMS

Try these division problems on your own!

$$1. 2/3 \div 1/4 = \underline{\hspace{2cm}}$$

$$2. 5/6 \div 2/3 = \underline{\hspace{2cm}}$$

$$3. 3/5 \div 1/2 = \underline{\hspace{2cm}}$$

$$4. 4/7 \div 2/5 = \underline{\hspace{2cm}}$$



KEY TAKEAWAYS

Remember these steps to divide fractions:

- Keep the first fraction the same
- Change $\div$ to $\times$ (division to multiplication)
- Flip the second fraction (reciprocal)
- Multiply across and simplify
- Practice makes perfect!



MATH IS FUN WHEN WE LEARN STEP-BY-STEP!

You've learned how to divide fractions like a superstar! Remember: Keep, Change, Flip, Multiply, and Simplify. With practice, you'll master fractions in no time. Believe in yourself!





**ANY QUESTIONS? LET'S KEEP
LEARNING TOGETHER!**

Thank you for learning with us today!